SAT Math Mastery · Session 6

Percentages, Ratios and Rates

Time 3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.

Topics percent of a quantity · percent increase and decrease · successive and reverse percent changes · percent error · mixtures and weighted percents · percent chains and parameters · ratios and proportions · rates and direct variation · unit conversion, including squared and cubed units · scale and relative speed · solving with a given total

Name / Date _____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. HARD · NO CALCULATOR NEEDED

The table below summarizes the number of students in each of four school clubs.

Club	Number of students
A	500
B	72
C	1,100
D	128
Total	1,800

The number of students in club C is $p\%$ of the number of students in club A. What is the value of p?

2. Hard · Desmos

In a factory quality check, 40% of the items inspected in a batch are defective. Of those defective items, 45% are also mislabeled. Which of the following is closest to the percentage of items in the batch that are NOT mislabeled defective items?

(A) 55%

(B) 82%

(C) 96.4%

(D) 18%

3. Hard · Desmos

The population of a town increased by 10% in the first year and then decreased by 10% in the second year. If the population at the start of the first year was 5,000, what was the population at the end of the second year?

(A) 5,050

(B) 4,900

(C) 4,950

(D) 5,000

4. Hard · Desmos

A renewable-energy report estimated that a new solar farm would generate 480 gigawatt-hours (GWh) of electricity in its first year. The farm actually generated 432 GWh. What is the percent error of the estimate, relative to the actual amount generated?

(A) 11.1%

(B) 12.5%

(C) 9.1%

(D) 10.0%

5. Hard · Desmos

The price of an item is increased by 40%, and then the new price is decreased by 25%. The final price is what percent of the original price?

(A) 115%

(B) 65%

(C) 100%

(D) 105%

6. Hard · Desmos

A store sells an item at 40% above its cost, then offers a 25% discount. The discounted selling price is \$210. What was the store's cost, in dollars?

7. Hard · Desmos

A town's population decreased by 20% in 2023 and then increased by 15% in 2024, ending at 9,200. What was the population before the 2023 decrease?

8. Hard · Desmos

The number x is 50% greater than y, and y is 20% less than z, where $z > 0$. The number x is what percent of z?

9. Hard · Desmos

A quantity increases by 10%, then increases by 10% again, and then decreases by 20%. By what percent has the quantity decreased from its original value? (Enter as a decimal.)

10. Hard · Desmos

A rectangular banner has an area of 540 square inches. The length of the banner is 150% of its width, and the width is $c\sqrt{10}$ inches, where c is a positive constant. What is the value of c?

11. Hard · Desmos

A metalworker melts together 150 g of alloy X, which is 20% nickel by mass, and m g of alloy Y, which is 50% nickel by mass. Then 40 g of pure nickel is stirred into the melt. The final alloy is 40% nickel by mass. What is the value of m?

(A) 250

(B) 460

(C) 60

(D) 100

12. Hard · Desmos

The area covered by a coral colony increases by 33.1% every 3 years. If the colony grows by the same percent each year, by what percent does its area increase each year?

13. Hard · Desmos

Last year a shop sold only mugs and plates. This year the number of mugs sold rose by 30%, the number of plates sold fell by 10%, and the total number of items sold rose by 6%. Of the items sold this year, what percent were mugs, to the nearest tenth of a percent?

(A) 40.0%

(B) 49.1%

(C) 50.9%

(D) 52.0%

14. Hard · Desmos

In 2020, the population of city M was 20% greater than the population of city N. From 2020 to 2024, the population of city M increased by 15% and the population of city N decreased by 8%. The population of city M in 2024 was p% greater than the population of city N in 2024. What is the value of p?

15. Hard · Desmos

The positive quantities a, b, c and d satisfy the following: b is 0.6% of a, c is 5% of b, and d is 150% of c. If $b + c + d = 27$, what is the value of a?

16. Hard · Desmos

A retailer's total sales were the same in March and April. In March, online sales were p percent of the total sales and came to \$18,000. In April, online sales were $(p + 8)$ percent of the total sales and came to \$25,200. What were the retailer's total sales, in dollars, for one of these months?

17. Hard · Desmos

A chemist has x liters of an acid solution that is 30% acid and mixes in 15 liters of pure acid. She then adds 25 liters of a second acid solution that is 10% acid. The final solution is 35% acid. What is the total volume, in liters, of the final solution?

(A) 125

(B) 70

(C) 85

(D) 110

18. Hard · Desmos

A library's collection consists of fiction, nonfiction and reference books. Next year the number of fiction books will increase by 20%, the number of nonfiction books will decrease by 10%, and the number of reference books will not change. After these changes, the collection will contain 1% more books than it does now, and the number of fiction books will equal the number of nonfiction books. What percent of the current collection is reference books?

(A) 65%

(B) 15%

(C) 20%

(D) 35%

19. Hard · Desmos

A store raises the price of a jacket by r percent. Later, it reduces the new price by $(r - 13)$ percent. After both changes, the final price is 10% higher than the original price. What is the value of r?

20. HARD · DESMOS

Quantity Q is 60% greater than quantity P. Quantity R is 25% less than Q. Quantity R is 50% greater than quantity S. P is what percent greater than S?

21. HARD · DESMOS

An 84-gram sample of an alloy contains only copper, zinc and tin. The mass of the copper is 5 grams more than twice the mass of the zinc. The mass of the tin is half the combined mass of the copper and the zinc. What is the mass, in grams, of the tin in the sample?

22. HARD · DESMOS

For a construction project, a concrete mixture is made from cement, sand, and gravel in the ratio $1 : 2 : 3$ by weight. A batching plant produces gravel at a constant rate of 6 metric tons per hour. At this rate, how many hours will it take the plant to produce 90 metric tons of the mixture?

23. Hard · Desmos

Car A travels 240 kilometers in 3 hours at a constant speed, and Car B travels 350 kilometers in 4.5 hours at a constant speed. Which of the following is closest to how much faster, in kilometers per hour, Car A's speed is compared with Car B's speed?

(A) 0.9

(B) 2.2

(C) 4.4

(D) 10.0

24. Hard · No calculator needed

A metal block has a mass of 540 grams and a volume of 60 cubic centimeters. What is the density of the block, in kilograms per cubic meter? (1 gram per cubic centimeter = 1,000 kilograms per cubic meter.)

(A) 90

(B) 900

(C) 9,000

(D) 9

25. Hard · Desmos

A water treatment plant produces fresh water at a constant rate of 1,250,000 liters per hour. At this rate, how many cubic meters of water does the plant produce in one day? (1 cubic meter = 1,000 liters.)

26. Hard · No calculator needed

The value of y varies directly with x. When $x = 9$, $y = 45$. What is the value of y when $x = 14$?

(A) 70

(B) 77

(C) 56

(D) 63

27. Hard · No calculator needed

A bus rapid transit line carries passengers at a constant average rate. The line carries 15,000 passengers over a 6-hour morning service window. At this same constant rate, which of the following is closest to the number of passengers carried during a 2.5-hour afternoon service window?

(A) 7,500

(B) 3,750

(C) 5,000

(D) 6,250

28. Hard · Desmos

Two cars are racing at constant speeds around a circular track. Car A requires 12 seconds to travel once around the track, and Car B requires 20 seconds to travel once around the track. If Car A passes Car B at a certain moment, how many seconds will elapse before Car A next passes Car B?

29. HARD · NO CALCULATOR NEEDED

A pump fills a rectangular tank at a rate of 0.8 cubic meter per minute. The tank is 12 meters long, 5 meters wide and 2 meters deep. How many hours does it take to fill the tank completely? (Enter as a decimal.)

30. HARD · DESMOS

A test vehicle accelerates at a constant 22 feet per second squared. Which of the following is equal to this acceleration in miles per minute squared? (1 mile = 5280 feet)

(A) $\frac{1}{4}$

(B) $\frac{1}{240}$

(C) 15

(D) 79,200

31. HARD · DESMOS

A car's fuel economy is 40 miles per gallon. Use 1 mile = 1.6 kilometers and 1 gallon = 3.8 liters. How many liters of fuel does the car need to drive 304 kilometers?

32. HARD · DESMOS

A region has a population density of 1,250 people per square kilometer. Using 1 mile = 1.6 kilometers, how many people are there per square mile?

33. HARD · DESMOS

A table has 4 legs, each with the same ground-contact area of A square centimeters. With a box on it, the table presses on the floor with a pressure of 12 newtons per square centimeter on each leg. A 120-newton weight is then added to the box, and the table is tilted so that only 3 legs touch the floor. The pressure on each of those 3 legs is now 24 newtons per square centimeter. (Pressure is force divided by contact area, and the load is shared equally by the legs touching the floor.) What is the value of A?

34. HARD · DESMOS

A box contains only red, green and blue flash drives, 140 in all. The ratio of red drives to green drives is 7 to 4. The ratio of blue drives to drives that are not blue is 3 to 11. How many more red drives than blue drives are in the box?

35. HARD · DESMOS

A site plan was drawn so that 1 centimeter on the plan represents 40 meters of ground. A copier then enlarged the plan to 125% of its size. On the enlarged copy, the parcel appears as a rectangle measuring 7.5 centimeters by 5 centimeters. What is the actual area of the parcel, in hectares? (1 hectare = 10,000 square meters.)

(A) 384

(B) 3.84

(C) 6

(D) 38.4

PART B · SELF-ASSESSMENT — 22 PROBLEMS

36. Medium · No calculator needed

The number m is 24% greater than 75. If m is the product of 75 and r, what is the value of r?

37. Medium · Desmos

A household's monthly electricity bill was \$340.00. After a rate adjustment, the bill is now \$378.20. To the nearest tenth of a percent, by what percent did the amount of the bill increase?

(A) 11.9%

(B) 12.4%

(C) 10.1%

(D) 11.2%

38. Medium · Desmos

Marcus purchased a car for \$90,000. He estimates that each year the value of the car will depreciate by 15% of the previous year's estimated value. What is the estimated value of the car exactly 2 years after Marcus purchased it?

(A) \$65,025

(B) \$76,500

(C) \$61,200

(D) \$63,750

39. Medium · Desmos

A school's enrollment decreased from 720 students to 640 students over one year. What is the percent decrease in enrollment?

(A) 10%

(B) 11.1%

(C) 12.5%

(D) 14.3%

40. Medium · Desmos

A laptop has a listed price of \$3,200. A store offers a 15% discount, and then a 15% sales tax is added to the discounted price. What is the final price of the laptop, in dollars?

(A) 2,976

(B) 3,008

(C) 3,128

(D) 3,200

41. Medium · No calculator needed

After a 20% discount, the price of a jacket is \$256. What was the original price, in dollars, of the jacket before the discount?

42. Medium · Desmos

A survey of 850 eleventh-grade students found that 34% plan to take the SAT this year. Which of the following is closest to the number of students who plan to take the SAT?

(A) 289

(B) 306

(C) 255

(D) 272

43. Medium · Desmos

If 30 percent of 40 percent of x is equal to 50 percent of y minus 20 percent of y, what is the value of x in terms of y?

(A) $x = 3y$

(B) $x = 5y$

(C) $x = 2y$

(D) $x = 2.5y$

44. Medium · Desmos

The mass of object A is k grams. The mass of object B is 130% of the mass of object A. If the mass of object B is 66 grams greater than the mass of object A, what is the value of k?

45. Medium · Desmos

A lab combines 40 grams of a solution that is 0.3% sodium chloride by mass with 60 grams of a solution that is 0.05% sodium chloride by mass. What is the percent of sodium chloride by mass in the combined 100 grams of mixture?

46. Medium · Desmos

A car consumes fuel at a rate of 6.5 liters per 100 kilometers. At this rate, how many liters of fuel are needed to travel 340 kilometers?

(A) 20.4

(B) 21.0

(C) 22.1

(D) 23.9

47. Medium · No calculator needed

A recipe for granola bars requires oats, honey, and raisins in the ratio 5 : 2 : 3 by weight. If a bakery uses 18 kilograms of raisins to prepare a large batch, how many kilograms of oats are used?

(A) 24

(B) 30

(C) 36

(D) 12

48. Medium · No calculator needed

A car travels at a constant speed of 90 kilometers per hour. What is this speed in meters per second?

(A) 32.4

(B) 324

(C) 15

(D) 25

49. Medium · No calculator needed

A recipe that serves 8 people uses 3 cups of orange juice and 5 cups of water. If the same ratio of ingredients is used to make a batch that serves 24 people, how many cups of orange juice are needed?

(A) 9

(B) 15

(C) 6

(D) 7.5

50. Medium · No calculator needed

A factory produces cardboard boxes at a constant rate of 480 boxes every 8 hours. At this rate, how many boxes does the factory produce in 5.5 hours?

51. Medium · No calculator needed

An architect built a scale model of a planned tower, where every 1 centimeter on the model represents 2.5 meters of the actual tower. If the model is 18 centimeters tall, what is the height, in meters, of the actual tower?

(A) 36

(B) 45

(C) 50

(D) 20

52. Medium · No calculator needed

A saline solution is mixed so that the ratio of salt to water is 2 to 15. If a container holds 51 liters of the solution in total, how many liters of salt are in the container?

(A) 12

(B) 3

(C) 6

(D) 9

53. Medium · Desmos

A gift box contains almonds and cashews, in a ratio of 7 to 5. When 8 almonds and 8 cashews are removed from the box, the ratio of almonds to cashews becomes 3 to 2. How many cashews were originally in the box?

(A) 40

(B) 48

(C) 24

(D) 32

54. Medium · Desmos

A rectangular banner is 2.4 meters by 1.5 meters. Printing costs \$0.05 per square centimeter. What is the cost, in dollars, to print the banner? (1 meter = 100 centimeters.)

55. Medium · Desmos

A train travels at a constant speed of 45 miles per hour. What is the train's speed, in feet per second? (1 mile = 5280 feet)

(A) 3,960

(B) 237,600

(C) 12.5

(D) 66

56. Medium · No calculator needed

A rectangular storage box is 0.4 meter long, 0.5 meter wide and 0.3 meter tall. What is the volume of the box, in cubic centimeters? (1 meter = 100 centimeters.)

57. Hard · Desmos

A rectangular banner has width w and length $1.8w$, where $w > 0$. The area of the banner equals the combined area of 4 identical square tiles. The side length of each tile is $k\,w\sqrt{5}$, where k is a positive constant. What is the value of k?